

Alexander Berezhnoy

Education

Saint Petersburg State University

September 2022 – June 2026

Bachelor's Degree in Applied Mathematics and Computer Science

- Faculty of Applied Mathematics and Control Processes
- Department of Information Systems
- Research: Clustering Methods Based on Satellite Imagery
- Strong command of numerical methods and optimization algorithms

ITMO University

September 2026 – June 2028 (in progress)

Master's Degree in Applied Computer Science

- Track: Frontend & Backend Development and AI Solutions

Skills

Languages: Russian, English

Technical: Python, C++, TypeScript, SQL, Git & GitHub, Java & Android Studio, Next.js

Projects & Experience

Numerical Methods (university coursework)

Implemented a set of numerical algorithms in Python: construction of interpolation and approximation polynomials, the Least Squares Method, solving ordinary differential equations using Runge-Kutta methods, and computing matrix eigenvalues via the power method. Used NumPy and Eigen for computations and matrix operations.

Frontend Developer — Clusterlab

Project developed with support from the Innovation Promotion Fund under the "Student Startup" program.

- Developed the client side of a precision agriculture web service using Next.js, React, and TypeScript: data-entry forms for field and soil information, interactive maps of vegetation indices, and fertilizer recommendation outputs.
- Integrated the frontend with a Python REST backend API to retrieve processed hyperspectral satellite imagery and recommendations.
- Contributed to backend and DevOps tasks: basic environment setup, build and deployment configuration.
- Collaborated within a development team: code review, branch and commit management in Git/GitHub.
- Used third-party services (Vercel, Railway, and Timeweb) for deployment, given the small scale of the project.